Part A - Assignment Based Questions

Q1. From your analysis of the categorical variables in the dataset, what could you infer about their effect on the dependent variable? (3 Marks)

I plotted boxplots of `cnt` against each categorical variable during EDA. Based on that, I observed the following:

- **season** – Demand is highest in fall, followed by summer and winter, while spring has the lowest demand. This suggests spring negatively affects bike rentals.
- **yr** – Demand in 2019 is clearly higher than in 2018. This indicates the bike-sharing service is becoming more popular year by year. So `yr` is an important predictor.
- **mnth** – Rentals are high from May to October and lower during December to February. This matches seasonal trends.
- **weathersit** – Demand decreases as weather becomes worse: clear weather > mist > light rain/snow. Poor weather strongly reduces demand.
- **holiday** – Holidays show slightly lower median demand compared to non-holidays.
- **weekday** – Most weekdays have similar demand levels, though Saturday shows a slight increase.
- **workingday** – Working days and non-working days have similar medians, but working days have a more consistent spread.

Conclusion:

Categorical variables such as `season`, `weathersit`, `yr`, and some months significantly affect bike demand. Therefore, they should be converted into dummy variables before applying linear regression.

Q2. Why is it important to use `drop_first=True` during dummy variable creation? (2 Marks)

When a categorical variable has **n categories**, one-hot encoding creates **n dummy columns**. These columns always add up to 1, meaning one column can be perfectly predicted from the others.

This creates the **dummy variable trap**, which causes perfect multicollinearity. In such cases, linear regression cannot calculate unique coefficients properly.

Using `drop_first=True` removes one dummy column, so only **n - 1 columns** remain.

Benefits:

- Removes perfect multicollinearity
- Prevents infinite VIF values

- Makes the model simpler
- Helps interpret coefficients relative to the dropped baseline category

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Q3. Looking at the pair plot among the numerical variables, which one has the highest correlation with the target variable? (1 Mark)

The variable **temp** has the highest correlation with `cnt`, approximately **0.63**.

`atemp` also has a similar correlation, but since it is almost identical to `temp` (correlation around 0.99), it was dropped during modelling to avoid multicollinearity.

Q4. How did you validate the assumptions of Linear Regression after building the model on the training set? (3 Marks)

I checked the main assumptions of linear regression as follows:

1. Linearity

I plotted residuals vs fitted values. The points were randomly scattered around zero with no clear curve, which supports a linear relationship.

2. Normality of Residuals

I used a histogram with KDE and a Q-Q plot. The residuals were approximately bell-shaped, and most Q-Q points were close to the diagonal line.

3. Homoscedasticity

The residuals vs fitted plot did not show any funnel shape, meaning variance was roughly constant.

4. No Multicollinearity

I calculated VIF values for all predictors. All were below 10 (highest around 7.7 for `temp`), so multicollinearity was acceptable.

5. Independence of Errors

Durbin-Watson statistic was around **2.06**, which is close to 2, indicating no autocorrelation.

Q5. Based on the final model, which are the top 3 features contributing significantly towards explaining the demand of shared bikes? (2 Marks)

Since numerical variables were MinMax scaled, coefficient values can be compared directly.

Top 3 significant features are:

1. **temp (+0.4039)**
   Temperature is the strongest positive factor. Warmer days increase rentals.
2. **weathersit_light_rain_snow (-0.2787)**
   Bad weather strongly reduces demand.
3. **yr (+0.2418)**
   Demand in 2019 is much higher than in 2018, showing yearly growth.

Part B - General Subjective Questions

Q1. Explain the Linear Regression algorithm in detail. (4 Marks)

Linear Regression is a supervised learning algorithm used to predict a continuous numerical value.

It assumes that the target variable depends linearly on one or more input variables.

Forms of Linear Regression:

Simple Linear Regression:

y = b0 + b1x + e

(One predictor)

Multiple Linear Regression:

y = b0 + b1x1 + b2x2 + ... + bnxn + e

(Multiple predictors)

Where:

- **b0** = intercept
- **b1, b2...bn** = coefficients
- **e** = error term

Training Method:

Linear regression mainly uses **Ordinary Least Squares (OLS)**, which finds coefficient values that minimise:

Sum of (Actual Value - Predicted Value)^2

Assumptions:

- Linear relationship
- Independent observations
- Constant variance of residuals
- Normally distributed residuals
- Low multicollinearity

Evaluation Metrics:

- R^2 Score
- Adjusted R^2
- MSE
- RMSE
- MAE

Advantages:

- Easy to understand
- Fast training
- Highly interpretable

Limitations:

- Cannot model complex non-linear data directly
- Sensitive to outliers
- Affected by multicollinearity

Q2. Explain Anscombe's Quartet in detail. (3 Marks)

Anscombe's Quartet is a famous example introduced by statistician **Francis Anscombe (1973)**.

It contains four different datasets that have almost identical statistical summaries but very different visual patterns.

Same Summary Statistics:

- Mean of x ≈ 9
- Mean of y ≈ 7.5
- Same variance
- Same correlation (~0.816)
- Same regression line

But Different Graphs:

1. **Dataset I** – Normal linear relationship
2. **Dataset II** – Curved relationship
3. **Dataset III** – One outlier disturbs regression
4. **Dataset IV** – One extreme point controls slope

Importance:

It teaches that summary statistics alone are not enough. Data should always be visualised before modelling.

Q3. What is Pearson's R? (3 Marks)

Pearson's correlation coefficient (R) measures the strength and direction of a **linear relationship** between two continuous variables.

Formula:

R = Cov(X,Y) / (Std(X) × Std(Y))

Range:

- **+1** = Perfect positive correlation
- **-1** = Perfect negative correlation
- **0** = No linear correlation

Interpretation:

- |R| < 0.3 = Weak
- 0.3 to 0.7 = Moderate
- 0.7 = Strong

Important Notes:

- Only measures linear relationships
- Sensitive to outliers
- Correlation does not imply causation

Q4. What is Scaling? Why is scaling performed? Difference between Normalisation and Standardisation? (3 Marks)

Scaling:

Scaling means converting numerical variables to a common range.

Why Scaling is Needed:

- Helps compare coefficients
- Improves performance of KNN, SVM, PCA, K-Means, Neural Networks
- Helps gradient descent converge faster
- Important in Ridge/Lasso regression

1. Normalisation (Min-Max Scaling)

x' = (x - min) / (max - min)

- Converts values to range [0,1]
- Sensitive to outliers

2. Standardisation (Z-score Scaling)

x' = (x - mean) / std deviation

- Mean becomes 0
- Standard deviation becomes 1
- Better when data is approximately normal

Q5. Why does VIF sometimes become infinite? (3 Marks)

Formula:

VIF = 1 / (1 - R^2)

If **R^2 = 1**, denominator becomes zero, so VIF becomes infinite.

This means one predictor can be perfectly predicted using other predictors.

Reasons:

- Dummy variable trap
- Duplicate columns
- Exact derived variables
- Too many features compared to rows

Solution:

Remove redundant variables or drop one dummy category.

Q6. What is a Q-Q Plot? Explain its use in Linear Regression. (3 Marks)

A **Q-Q Plot (Quantile-Quantile Plot)** compares sample data distribution with a theoretical normal distribution.

Interpretation:

- Points on straight line → approximately normal
- Curved ends → heavy/light tails
- S-shape → skewness
- Extreme deviations → outliers

Importance in Linear Regression:

- Checks whether residuals are normally distributed
- Helps validate OLS assumptions
- Useful for checking p-values and confidence intervals
- Detects outliers and skewness

Conclusion:

Q-Q plot is an important diagnostic tool used after fitting regression models.